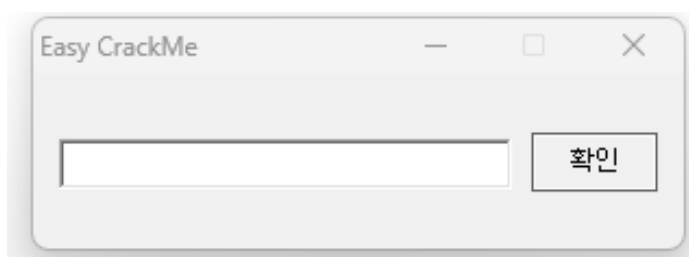




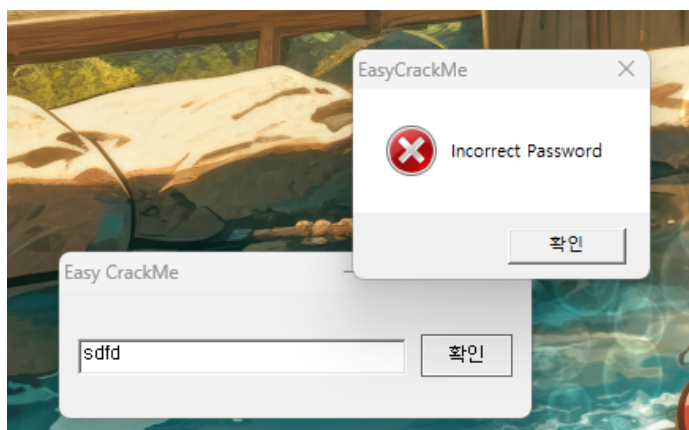
1. Easy Crack Me

Easy_CrackMe.zip

문제 파일 다운로드

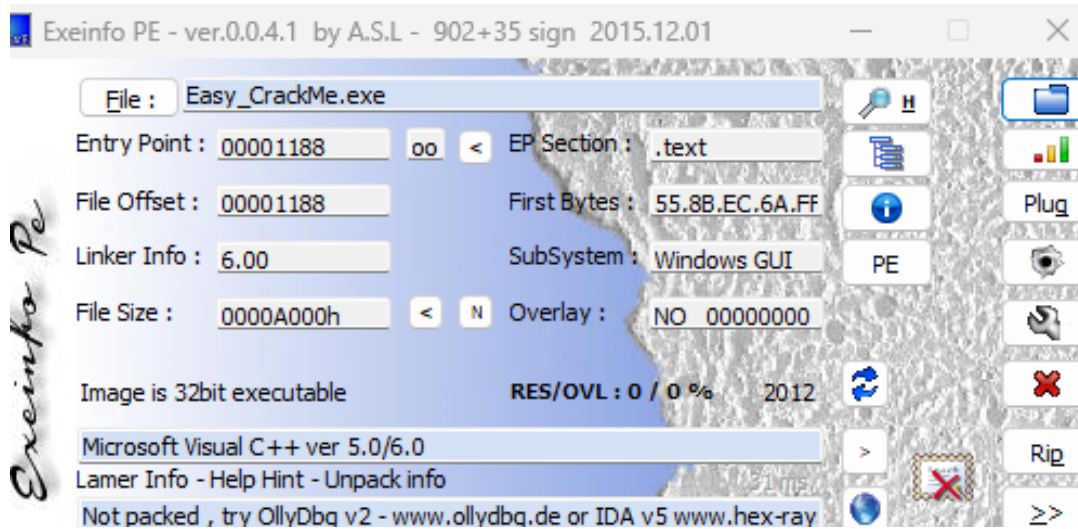


- 암호 입력 시 통과 여부를 알려줌



분석

- x32 Debugger
- Exeinfo PE



- 패킹의 여부를 확인

00401000	8B4424 04	mov eax,dword ptr ss:[esp+4]
00401004	6A 00	push 0
00401006	68 20104000	push easy_crackme.401020
0040100B	6A 00	push 0
0040100D	6A 65	push 65
0040100F	50	push eax
00401010	FF15 A8504000	call dword ptr ds:[<DialogBoxParamA>]
00401016	33C0	xor eax,eax
00401018	C2 1000	ret 10
0040101B	90	nop

- 문자열 참조부분 상단에서 Dialog BoxParmA 함수를 호출하는 것을 확인

004010A3	50	push eax	
004010A4	68 E8030000	push 3E8	
004010A9	57	push edi	
004010AA	FF15 9C504000	call dword ptr ds:[<GetDlgItemTextA>]	
004010B0	807C24 05 61	cmp byte ptr ss:[esp+5],61	61: 'a'
004010B5	75 7E	jne easy_crackme.401135	
004010B7	6A 02	push 2	
004010B9	8D4C24 0A	lea ecx,dword ptr ss:[esp+A]	
004010BD	68 78604000	push easy_crackme.406078	406078: "5y"
004010C2	51	push ecx	
004010C3	E8 88000000	call easy_crackme.401150	
004010C8	83C4 0C	add esp,C	
004010CB	85C0	test eax,eax	
004010CD	75 66	jne easy_crackme.401135	
004010CF	53	push ebx	
004010D0	56	push esi	
004010D1	BE 6C604000	mov esi,easy_crackme.40606C	40606C: "R3versing"
004010D6	8D4424 10	lea eax,dword ptr ss:[esp+10]	

- GetDlgItemTextA 함수 호출

cmp byte ptr ss:[esp+5], 61 ;a (두번째 입력값)

esp 스택은 4바이트씩 할당되므로 첫번째 값이 아니라 2번째 값을 의미하며 61은 아스키 코드로 a를 의미 ⇒ 두번째 입력값은 a인것을 확인

jne easy_crackme.401135 ; 두번째 문자열이 a가 아닐경우 incorrect password 주소로 점프

• 현재 알아낸 정보 X a XXXXX

push easy_crackme.406078 ; "5y"
call easy_crackme.401150 ; easy crackme 호출

• 5y라는 문자열을 확인

004010A9	57	push edi	
004010AA	FF15 9C504000	call dword ptr ds:[<GetDlgItemTextA>]	
004010B0	807C24 05 61	cmp byte ptr ss:[esp+5],61	61: 'a'
004010B5	75 7E	jne easy_crackme.401135	
004010B7	6A 02	push 2	
004010B9	8D4C24 0A	lea ecx,dword ptr ss:[esp+A]	
004010BD	68 78604000	push easy_crackme.406078	406078: "5y"
004010C2	51	push ecx	ecx: "5y"
004010C3	E8 88000000	call easy_crackme.401150	

• ecx에 입력한 값을 넣어 5y와 비교

00401164	F2:AE	repne scasb	
00401166	F7D9	neg ecx	ecx: "5y"
00401168	03CB	add ecx,ebx	ecx: "5y", ebx: "5y"
0040116A	8BFE	mov edi,esi	
0040116C	8B75 0C	mov esi,dword ptr ss:[ebp+C]	[ebp+0C]: "5y"
0040116F	F3:A6	repe cmpsb	
00401171	8A46 FF	mov al,byte ptr ds:[esi-1]	
00401174	33C9	xor ecx,ecx	ecx: "5y"
00401176	3A47 FF	cmp al,byte ptr ds:[edi-1]	
00401179	77 04	ja easy_crackme.40117F	
0040117B	74 04	je easy_crackme.401181	
0040117D	49	dec ecx	ecx: "5y"
0040117E	49	dec ecx	ecx: "5y"
0040117F	F7D1	not ecx	ecx: "5y"
00401181	8BC1	mov eax,ecx	ecx: "5y"
00401183	5B	pop ebx	ebx: "5y"
00401184	5E	pop esi	
00401185	5F	pop edi	
004010CF	53	push ebx	ebx: "5y"
004010D0	56	push esi	
004010D1	BE 6C604000	mov esi,easy_crackme.40606C	40606C: "R3versing"
004010D6	8D4424 10	lea eax,dword ptr ss:[esp+10]	
004010DA	8A10	mov dl,byte ptr ds:[eax]	
004010DC	8A1E	mov bl,byte ptr ds:[esi]	
004010DE	8ACA	mov cl,dl	
004010E0	3AD3	cmp dl,bl	
004010E2	75 1E	jne easy_crackme.401102	

• 현재 알아낸 정보 X a 5yXXX

mov esi, easy_crackme.40606c ; "R3versing"
lea eax, dword ptr ss:[esp+10]
mov dl, byte ptr ds:[eax] ; 입력값
mov bl, byte ptr ds:[esi] ; bl = R3versing = 현재 esi에 들어있는 값
mov cl, dl ; cl == 사용자가 입력한 값
cmp dl,bl ; 사용자가 입력한 값 과 R3versing 값과 비교
jne easy_crackme.401102 ; 값이 다르면 Incorrect password로 점프

- 현재 알아낸 정보 X a5yR3versing

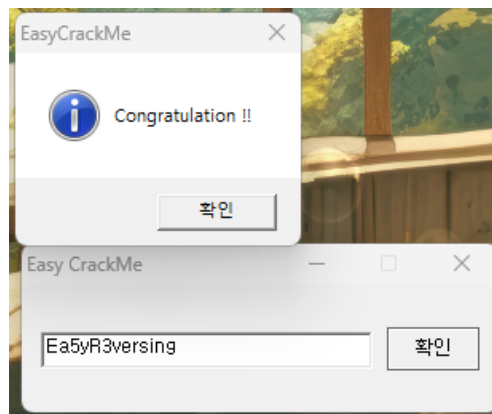
00401108	5B	pop ebx	ebx: "y"
00401109	85C0	test eax, eax	
0040110B	75 28	jne easy_crackme.401135	
0040110D	807C24 04 45	cmp byte ptr ss:[esp+4], 45	45: 'E'
00401112	75 21	jne easy_crackme.401135	
00401114	6A 40	push 40	
00401116	68 58604000	push easy_crackme.406058	406058: "EasyCrackMe"
00401118	68 44604000	push easy_crackme.406044	406044: "Congratulation !!"

```

pop ebx
test eax, eax
jne easy_crackme.401135 ; incorrect password
cmp byte ptr ss:[esp+4], 45 ; 45 = 'E'

```

- 이전에 알아낸 a는 esp + 5에 저장되어 두번째 값을 의미
- 현재 E라는 값은 esp + 4에 저장되어 있으므로 첫번째 값을 의미
- 현재 알아낸 정보 Ea5yR3versing



FLAG : Ea5yR3versing

<https://learn.microsoft.com/en-us/windows/win32/api/winuser/nf-winuser-dialogboxparama>

<https://learn.microsoft.com/ko-kr/windows/win32/api/winuser/nf-winuser-getdlgitemtexta>